

MICROBIOLOGY – SHORT EXAM NOTES

Components of the Immune System (Ananthanarayan & Paniker, Ch. 14 – Structure and Functions of the Immune System)

1. INTRODUCTION

The **lymphoreticular system** is a complex organisation of cells distributed in different organs and tissues, responsible for immunity. It has two components: the **lymphoid cells** (lymphocytes and plasma cells), concerned with the **specific immune response**; and the **reticuloendothelial (phagocytic) cells**, which have '**scavenger**' functions – removing effete cells, foreign particles and microorganisms (non-specific immunity) – and also help in specific immunity.

Types of immune response: (1) **Humoral (antibody-mediated) immunity** – by antibodies produced by **plasma cells** in blood and body fluids ('humor'); (2) **Cellular (cell-mediated) immunity** – directly by **sensitised T lymphocytes**. The two develop through separate channels but interact.

Hematopoiesis: all immune cells arise from the **hematopoietic stem cell (HSC)** in the bone marrow, which gives a **lymphoid progenitor** (T cells, B cells, NK cells, dendritic cells) and a **myeloid progenitor** (neutrophils, monocytes/macrophages, eosinophils, basophils, platelets, erythrocytes, dendritic cells).

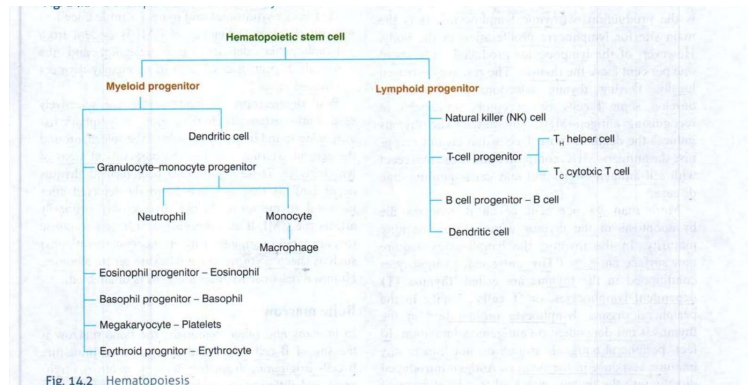


Fig. 14.2 Hematopoiesis

Fig. 14.2 Hematopoiesis (from the book)

The components of the immune system are therefore the **lymphoid organs**, the **lymphoid cells** and the **phagocytic and antigen-presenting cells**, together with molecules such as the **MHC**.

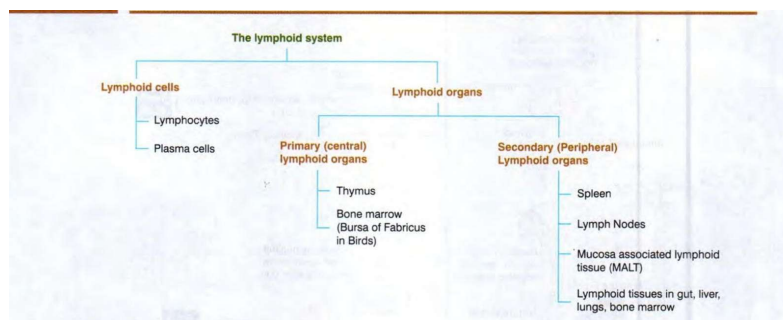


Fig. 14.3 The lymphoid system (from the book)

2. LYMPHOID ORGANS

A. Central (Primary) Lymphoid Organs

Sites where lymphocytes **develop and mature** (become immunocompetent); proliferation here is **not antigen-dependent**.

Thymus

- **Structure:** behind the upper sternum; **two lobes** in a fibrous capsule, divided into lobules with an outer **cortex** (densely packed proliferating lymphocytes) and inner **medulla** (epithelial cells, mature lymphocytes and **Hassall's corpuscles** – whorls of epithelial cells). Maximal relative size just before birth; grows till ~12 years, then **involutates after puberty**.
- **Function:** produces and 'educates' **T (thymus-dependent) lymphocytes**. Only about **1%** of lymphocytes produced leave the thymus; **>95% die by apoptosis** – positive selection keeps cells recognising antigen–MHC, negative selection removes self-reactive cells (**self-tolerance**). T cells acquire 'Thy' antigens and **immunocompetence** for cell-mediated immunity.
- **Clinical: DiGeorge syndrome** – congenital thymic aplasia with deficient CMI ('nude mice' in animals); **runt disease** after neonatal thymectomy in mice. Thymus-dependent areas: **paracortex of lymph nodes** and **periarteriolar sheath of spleen**.

Bone Marrow

The site of origin of **all lymphocytes** and of **B cell development** in humans (equivalent of the **bursa of Fabricius in birds**). Immature B cells proliferate and differentiate with the help of cytokines; self-reactive B cells are eliminated. The fetus produces IgM and IgD by the 20th week; maternal IgG is received; IgG production becomes adequate by 2–3 years; full immunocompetence after the first decade.

B. Peripheral (Secondary) Lymphoid Organs

Sites where mature lymphocytes **meet antigens** and mount immune responses.

Lymph Nodes

Placed along lymphatic vessels, with an outer **cortex** and inner **medulla**. The cortex has **primary lymphoid follicles** in which **germinal centres (secondary follicles)** develop on antigenic stimulation; the follicles and **medullary cords** contain **B cells** (bursa-dependent areas). The **paracortical area** between them contains **T cells** and interdigitating cells (**thymus-dependent area**). **Functions:** filter lymph from a specific region, phagocytose foreign material and microbes, help proliferation and circulation of T and B cells, and enlarge on local antigenic stimulation.

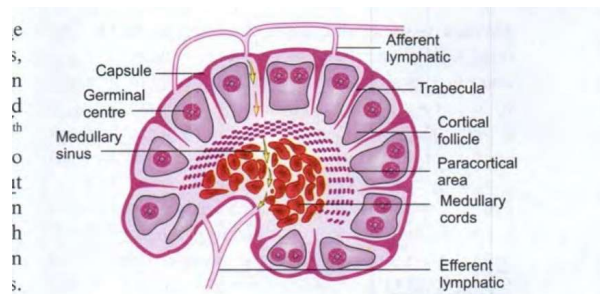


Fig. 14.4 Diagrammatic section of lymph node (from the book)

Spleen

The **largest lymphoid organ**; capsule with trabeculae; **white pulp** (lymphoid) and **red pulp**. The **periarteriolar lymphatic sheath** around the central arteriole is the **thymus-dependent area**; the follicles, germinal centres, mantle layer and perifollicular region are thymus-independent. **Functions:** graveyard for effete blood cells, blood reservoir, and a filter for **bloodborne antigens**. **Splenectomy** in children increases risk of sepsis by capsulated bacteria – *S. pneumoniae*, *N. meningitidis*, *H. influenzae*.

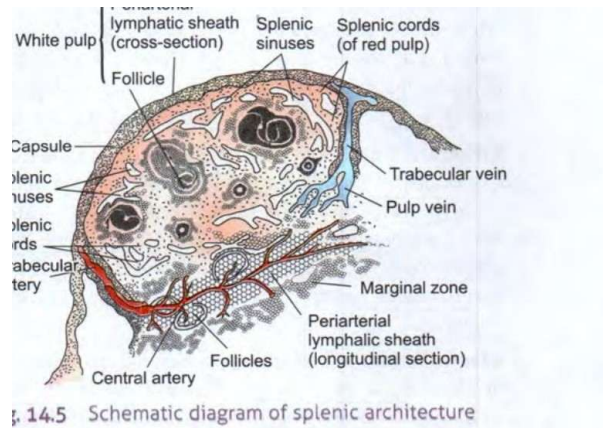


Fig. 14.5 Schematic diagram of splenic architecture (from the book)

Mucosa-Associated Lymphoid Tissue (MALT)

Lymphoid tissue lining the alimentary, respiratory and genitourinary tracts – scattered cells, **Peyer's patches**, tonsils, adenoids and follicles. It includes **GALT** (gut) and **BALT** (bronchus). MALT has more antibody-producing plasma cells than spleen, nodes and marrow together; the main antibody is **secretory IgA**. A **common mucosal immune system** explains the superiority of **oral/nasal immunisation** for enteric and respiratory infections.

Lymphocyte recirculation: lymphocytes (mainly **T cells**) move continuously through blood, lymph and lymphoid organs ('policeman on beat patrol'), completing a cycle in 1–2 days, so that antigen anywhere in the body meets the right lymphocytes.

3. CELLS OF THE IMMUNE SYSTEM

A. Lymphocytes

Small round cells forming **20–40% of WBCs** and **99% of cells in lymph**; about 10^{12} in the body. The main cells of the immune response – they **recognise antigen, store memory and respond specifically**. Types: **T cells, B cells and natural killer (NK) cells**. **Short-lived** lymphocytes (about 2 weeks) are effector cells; **long-lived** ones (years/life, mainly T cells) store **immunological memory**. A lymphocyte educated in a central organ is an **immunologically competent cell**; before meeting antigen it is a **naïve cell**.

CD markers: surface antigens identified by monoclonal antibodies and given a **cluster of differentiation (CD)** number, e.g. **CD3** (all T cells), **CD4** (helper T; HIV receptor), **CD8** (cytotoxic/suppressor T), **CD19** (B cells), **CD2** (SRBC receptor).

Differences between T cells and B cells

Feature	T cell	B cell
Site of maturation	Thymus	Bone marrow (bursa in birds)
Immunity	Cell-mediated	Humoral (antibodies via plasma cells)
Antigen receptor	TCR + CD3	Surface Ig (IgM, IgD)
Marker	CD3, CD4/CD8, CD2	CD19, surface Ig, C3 receptor (CR2 – EBV receptor)
Rosette	E rosette with sheep RBC (CD2)	EAC rosette (C3 receptor)
Mitogen	PHA, Concanavalin A	Endotoxin (LPS), <i>S. aureus</i> Cowan I, EB virus
Surface (SEM)	Smooth	Numerous microvilli

Feature	T cell	B cell
Recirculation	Mostly recirculating	More sessile
Location in lymph node	Paracortex	Follicles, medullary cords

T Cells

T cell maturation: precursors from yolk sac, fetal liver and bone marrow migrate to the thymus: **pro-T** → **pre-T** → **immature T** → **mature CD4+ or CD8+ T cells**, gaining CD7, CD2, CD3, TCR, CD4 and CD8. **T cell receptor (TCR)** – a heterodimer of $\alpha\beta$ (most T cells) or $\gamma\delta$ chains linked to **CD3**; has V, D, J and C regions (immunoglobulin gene superfamily). T cells develop **MHC restriction**: CD8+ cells recognise antigen with **HLA class I**, CD4+ cells with **HLA class II**.

Type	Marker / restriction	Function
Helper T (TH)	CD4, MHC class II	Stimulate B cells, T cells and macrophages
– TH1	CD4	IFN-γ, IL-2 ; activate macrophages; CMI against intracellular pathogens (TB, leprosy); delayed hypersensitivity
– TH2	CD4	IL-4, 5, 6 ; stimulate B cells to form antibodies
– TH17	CD4	IL-17 ; promote inflammation, autoimmunity
Regulatory T (Treg)	CD4	TGF-β ; regulate immune response, self-tolerance
Cytotoxic T (Tc/CTL)	CD8, MHC class I	Kill virus-infected, tumour and allograft cells
Memory T (Tm)	CD4 or CD8	Memory and anamnestic response
$\gamma\delta$ T cells	TCR $\gamma\delta$	Surveillance of epithelia; innate immunity

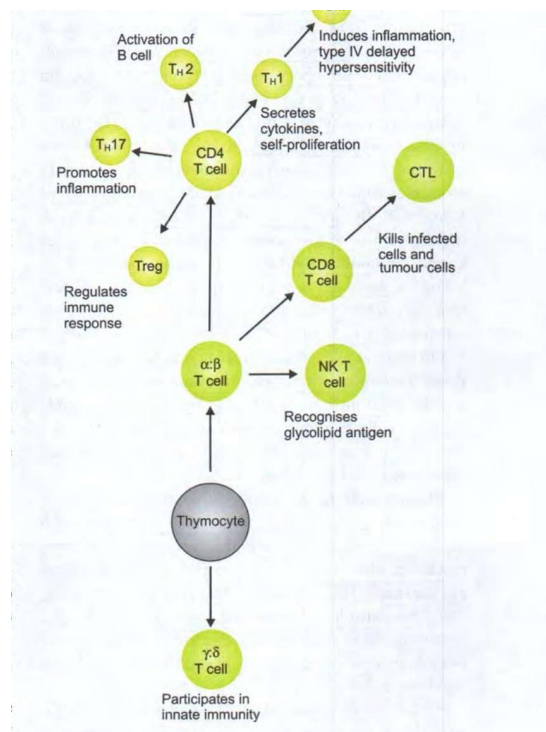


Fig. 14.8 Types of T cells and their functions (from the book)

B Cells and Plasma Cells

B cell maturation: stem cell → pro-B → pre-B (cytoplasmic IgM) → immature B (surface IgM) → naïve/mature B (surface IgM + IgD) in the bone marrow (antigen-independent), then antigen-dependent activation in the periphery. By **allelic exclusion** each B cell makes one class and one light chain type of a single specificity; self-tolerance by clonal deletion/anergy. On meeting antigen, the B cell undergoes **clonal proliferation** into **plasma cells** and long-lived **memory B cells** (secondary response). **CD5+ B-1 cells** secrete low-affinity 'natural' IgM.

Plasma cell – the **antibody-secreting cell**; oval, twice the size of a lymphocyte, with an **eccentric nucleus** showing '**cartwheel**' **chromatin**, abundant endoplasmic reticulum and Golgi; an end cell living **2–3 days**; makes antibody of a single specificity, class and light-chain type.

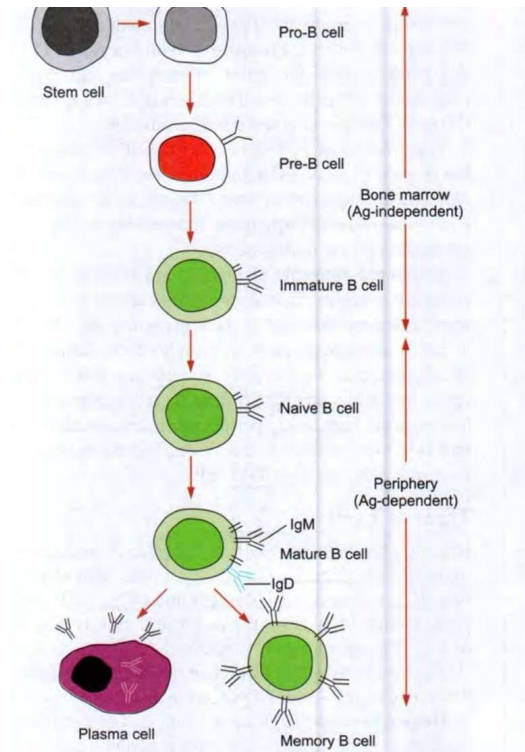


Fig. 14.9 B cell maturation (from the book)

Null Cells (Large Granular Lymphocytes)

5–10% of lymphocytes lack T and B markers; large granular lymphocytes (LGL).

- **Natural killer (NK) cells** – spontaneous cytotoxicity against **virus-infected and tumour cells**, **not antibody-dependent or MHC-restricted**, no prior sensitisation (part of **innate immunity**); markers **CD16, CD56**; release **perforin** (resembles C9) causing apoptosis; enhanced by interferon.
- **Antibody-dependent cytotoxic (ADCC) cells** (former K cells) – Fc receptors; kill IgG-coated targets.
- **Lymphokine-activated killer (LAK) cells** – NK cells treated with **IL-2**; kill tumour cells (renal cell carcinoma).

B. Phagocytic Cells

Phagocytosis is the oldest defence mechanism. Two types:

- **Mononuclear phagocytes (macrophages):** blood **monocytes** (12–15 μm , half-life ~ 3 days) become tissue **macrophages (histiocytes)** (15–20 μm , live months): **alveolar macrophages**, **Kupffer cells**, **osteoclasts**, **mesangial cells**, **microglia**; epithelioid and giant cells in granulomas. **Functions:** phagocytosis, **antigen processing and presentation** (MHC restricted), secretion of **IL-1** (endogenous

pyrogen), TNF, CSF, enzymes; **activated** by lymphokines, complement or interferon; 'armed' macrophages show antitumour activity. Some organisms (typhoid, brucella, TB bacilli) survive inside them.

- **Polymorphonuclear microphages: neutrophils** – actively phagocytic, main cells of **acute inflammation**; **eosinophils** – allergy and **parasitic infections**, extracellular killing of large parasites; **basophils/mast cells** – granules with **histamine, heparin, serotonin**; degranulation causes **anaphylaxis and atopy**.

C. Antigen-Presenting Cells (APCs)

TH cells recognise antigen only when displayed with **MHC class II** on APCs – **dendritic cells, macrophages and B cells**. **Dendritic cells** are the '**professional**' APCs – bone-marrow derived, with long processes, class II MHC and co-stimulatory molecules (B7, CD28); key in the **primary response**. **B cells** present antigen mainly in the **secondary response**. **Langerhans cells** in skin process antigens reaching the dermis.

Defects of immune cells (Table 14.3)

Defect	Disease
Stem cell differentiation	Neutropenia
Lack of CD18 – adhesion failure	Leukocyte adhesion deficiency
No phagosome–lysosome fusion	Chediak–Higashi syndrome
Defective NADPH oxidase – intracellular killing	Chronic granulomatous disease
Congenital thymic aplasia	DiGeorge syndrome (deficient CMI)

4. MAJOR HISTOCOMPATIBILITY COMPLEX (MHC)

A group of genes coding for cell-surface antigens responsible for **graft rejection**; discovered by **Gorer** (H-2 antigen in mice). In humans the MHC antigens are **human leukocyte antigens (HLA)**, and the gene complex lies on the **short arm of chromosome 6**. (Nobel Prize 1980 – Snell, Dausset, Benacerraf; 1996 – Doherty & Zinkernagel for MHC restriction.)

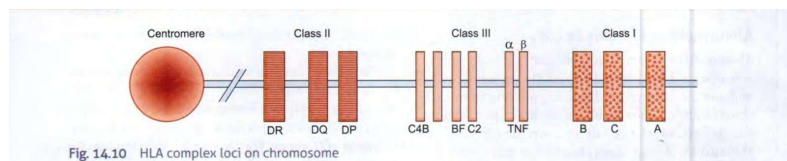


Fig. 14.10 HLA complex loci on chromosome 6 (from the book)

Feature	Class I	Class II	Class III
Loci	HLA-A, B, C	HLA-DR, DQ, DP (D region)	Complement C2, C4, factor B , TNF, heat shock proteins
Structure	Heavy α chain (3 domains) + β2-microglobulin (chromosome 15)	α and β chains (2 domains each)	Heterogeneous
Distribution	All nucleated cells	Immune cells – B cells, macrophages, dendritic cells, activated T cells	–
Presents to	CD8 cytotoxic T cells	CD4 helper T cells	–

Feature	Class I	Class II	Class III
Role	Graft rejection, killing of virus-infected cells	Regulate immune response; GVH reaction, mixed leukocyte reaction	Complement

HLA typing – serologically by **microcytotoxicity** (complement-mediated lysis of lymphocytes; antisera from multiparous women or monoclonals), MLR for DR, and molecular methods (RFLP, SSO probes).

Applications: donor–recipient matching in **transplantation**, **disputed paternity**, **anthropology**, and **disease association** – **HLA-B27** with ankylosing spondylitis, **DR4** with rheumatoid arthritis, **DR3** with autoimmune diseases. **MHC restriction:** T cells respond to antigen only when presented with **self-MHC**.

★ EXAM FOCUS

- Short answers: **lymphoid organs and cells, T vs B cells, types of T cells, types of immune response, MHC classes.**
- Short notes: **B cells, T cells, TCR, antigen-presenting cells, null cells, phagocytic cells, role of macrophages, HLA typing and applications, MHC restriction.**
- One-liners: primary organs – thymus & bone marrow; Hassall's corpuscles – thymus medulla; T-cell area – paracortex; B-cell mitogen – LPS; T-cell mitogens – PHA, Con A; NK markers – CD16, CD56; HLA on chromosome 6.